

Nikhil Roy

21niroy70@gmail.com | 412-327-6294 | [LinkedIn](#): nikhil-roy-453441221 | [GitHub](#): 21niroy70 | Pittsburgh, Pennsylvania | US Citizen

EDUCATION

University of Maryland, College Park, MD
Master of Science: Applied Machine Learning

Graduation: December 2025
GPA: 4.0

University of Pittsburgh, Pittsburgh, PA
Bachelor of Science: Data Science, Minor in Applied Statistics

Graduation: April 2024
GPA: 3.7

Skills: Python, SQL, R, Databricks, PySpark, Jupyter Notebook, Docker, AWS, NLP, Deep learning, Git, Statistics, Communication

PROFESSIONAL EXPERIENCE

General Motors

Warren, Michigan

Data Scientist Intern

May 2025 – August 2025

- Crafted a Product Engagement Score using Python, PySpark, SQL, and ML within Databricks, aggregating 3.5+ billion customer interactions across OnStar services (Remote Start, Wi-Fi, Super Cruise) into a unified Customer Engagement Score
- Deployed churn prediction and lifetime value (LTV) through Confirmatory Factor Analysis (CFA) methodology to boost customer retention and acquisition, delivering insights to 32 cross-functional stakeholders in marketing, sales, and finance
- Engineered predictive features through log transformations, exponential decay functions, and PCA, enhancing OnStar engagement models, reducing feature gaps by 37%, and contributing to a 6% revenue increase in Q2

UnitedHealth Group

Pittsburgh, PA

Statistical Data Scientist Intern

June 2024 – August 2024

- Applied advanced statistical and mathematical methods in R to formulate optimal predictive machine learning models, classifying poor mental health days, impacting over 1.5 billion patients across the United States, India, and Ireland
- Published an executive paper detailing value proposition and strategy to present to hundreds of end clients and senior leadership through the Bridges program hosted by Carnegie Mellon University

University of Pittsburgh

Pittsburgh, PA

Neurocognitive Machine Learning Researcher

July 2023 – August 2024

- Utilized mathematical unsupervised machine learning algorithms to discover trends with Neural Cognitive Systems (NCS), achieving 87% precision on identifying patterns dealing with brain behaviors through theoretical statistical analysis
- Optimized data pipelines for audio feature extraction, achieving a 30% reduction in data processing time while maintaining marginal performance

PERSONAL PROJECTS

Computer Vision Automated Vehicle Speed Detection

- Pioneered a real-time computer-vision pipeline using multi-class vehicle detection, achieving a 93% mean average precision (mAP) and ± 2 mph speed-estimation accuracy on 1080p KITTI dash-cam video at 30 FPS
- Automated CI/CD deployment with Docker containers orchestrated via Airflow and GitHub Actions; published a dashboard streaming live vehicle detection, speed monitoring, and speeding alerts, providing real-time insights for automobile clients

NASA Solar Flare Identification Data Scientist

- Collaborated with NASA's secure data hub to automate real-time web scraping of unstructured solar flare data utilizing BeautifulSoup and Selenium Webdriver, processing over 6 million coordinates weekly
- Enhanced detailed data tidying and preprocessing, including normalization and feature extraction, further applying machine learning techniques to identify clusters, resulting in a 36% efficiency in solar flare identification

Carnegie Mellon University (CMU) Machine Learning Engineer Research

- Developed advanced supervised learning algorithms to forecast crucial paint characteristics for a local Pittsburgh company
- Integrated Bayesian Statistical models and tuned advanced models from unstructured data, resulting in 35% reduction in business costs, 30% increase in cloud storage, and 20% improvement on accuracy through generating posterior predictions
- Evaluated and visualized performance metrics such as prediction intervals, predictive accuracy, RMSE, F1, and ROC to enhance business decisions and boost profits for distributions by over 140%

LEADERSHIP

Taekwondo

- Triumphed 4 gold medals in international sparring tournaments, breaking boards, and competitive use of weapons
- Taught dozens of evening classes to students needing assistance of self-defense, sparring, communication, and discipline

South Asian Student Association (SASA)

- Coordinated 2 shows with diverse dance performances, embracing cross-cultural appreciation for thousands of students
- Lead events aimed at supporting community-based organizations such as Glen Hazel and YMCA Urban Garden Projects